

Hetang Mehta

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EDUCATION

University of Alberta

April 2027

BSc Computer Engineering (Co-op Program)

Edmonton, AB

- Coursework: Machine Learning, Data Analytics, Parallel and Distributed Systems, Operating Systems, Data Structures and Algorithms, Computer Architecture, File and Database Management

EXPERIENCE

Machine Learning Engineer

Sep 2025 – Present

Alberta BIONIX - School Club

Edmonton, AB

- Processed and analyzed a **26GB EMG gait dataset** using **Python (Pandas, NumPy, Scikit-learn)**, building feature engineering pipelines to extract biomechanical signals used for prosthetic knee control.
- Developed and evaluated **machine learning models (Random Forest, SVM)** to classify prosthetic trial participant viability from EMG activation patterns.
- Designed and trained **CNN models for real-time knee flexion prediction**, achieving **87% accuracy** across locomotion modes (sitting, standing, walking, running).

Software Engineer Intern

Jan 2024 – Aug 2025

Canadian Natural Resources Limited (CNRL)

Calgary, AB

- Improved productivity by **30%** for **500+** engineers by owning the design and delivery of a PWA built with **C#, .NET, Blazor, REST APIs**, and **Azure CI/CD**.
- Reduced defects by **33%** and saved **\$40K** annually by redesigning **Oracle SQL** schemas, optimizing queries, and enforcing data integrity constraints.
- Reduced test latency from **10 days to 4 minutes**, saving **\$60K** by building **300+** automated unit tests and validation pipelines in **C#** and **VB**.

Software Engineer Intern

Sep 2023 – Dec 2023

Suncor Energy

Calgary, AB

- Developed Python-based automation scripts for SAP workflows, saving **\$26K** annually.
- Processed and cleaned large datasets (data preprocessing) using Python, Excel, and VB; built analytics dashboards in Power BI for decision-making.

PROJECTS

Event Lottery System | *Java, Android Studio, Firebase*

- Built an Android application in Java, enabling fair, lottery-based registration for high-demand community events.
- Integrated Firebase for real-time event data, notifications, user state management, and multi-role access control.
- Supported event life-cycle management, including registration windows, cancellations, and replacement draws.

ATS Resume Analyzer | *Demo* | *Next.js, TypeScript, React.Ts, Google Gemini AI*

- Developed a full-stack web application with **Next.js 15, React.ts, and TypeScript** to evaluate resume compatibility with job descriptions.
- Built a **GenAI-powered resume evaluation platform** using **Google Gemini LLMs** to analyze job descriptions and compute ATS compatibility scores.
- Implemented prompt-based evaluation pipelines for keyword extraction and resume optimization recommendations.

Boxed | *React.ts, Node.js, Express, Auth0 Cloudflare R2 (S3 Storage), RESTful APIs*

- Built a cloud storage infrastructure enabling secure file uploads, retrieval, and access control.
- Implemented authentication and role-based access with Auth0 (using the OAuth 2.0 protocol); deployed on Docker with S3-compatible storage (Cloudflare R2).

TECHNICAL SKILLS

Languages: Python, Java, C++, C, C#, JavaScript, SQL, Ruby on Rails, Go, Golang, Swift, VB.NET, Kotlin

ML / AI: PyTorch, Scikit-learn, Pandas, NumPy, Gemini API

Cloud/DevOps: AWS S3, Cloudflare R2, Azure (Pipelines, CI/CD), Docker, Git, GitHub

Web/Frameworks/Tools: React.js, Node.js, SpringBoot, Blazor, .NET, RESTful APIs, Flask, JUnit, Linux/Unix Shell, Auth0, Android Studio, Kubernetes

Databases: SQL (Oracle, Postgres, MySQL), NoSQL (MongoDB)